**PS#12 - Taylor polynomials - Answer key**

1. (AC 8a.2 Exercise 5) Suppose we know the following information about a function $f$:

$$f(0)=2, f'(0)=-3, f''(0)=-1, f'''(0)=0, f^{(4)}(0)=-3.$$

(a) Determine $T_4(x)$, the degree 4 Taylor polynomial of $f$ that is centered at $a=0$.

$$\begin{aligned} T_4(x) &= f(0)+\frac{f'(0)}{1!}x^1+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4 \\ &= 2+\frac{-3}{1!}x^1+\frac{-1}{2!}x^2+\frac{0}{3!}x^3+\frac{-3}{4!}x^4 \\ &= 2-3x-\frac{1}{2}x^2-\frac{1}{8}x^4 \end{aligned}$$

(b) Use $T_4(x)$ to estimate $f(0.5)$.
$f(0.5) \approx T_4(0.5) = 0.3671875.$

(c) State each of the Taylor polynomials $T_3(x)$, $T_2(x)$, and $T_1(x)$.

$$\begin{aligned} T_4(x) &= 2-3x-\frac{1}{2}x^2-\frac{1}{8}x^4 \\ T_3(x) &= 2-3x-\frac{1}{2}x^2(+0x^3) \\ T_2(x) &= 2-3x-\frac{1}{2}x^2 \\ T_1(x) &= 2-3x \end{aligned}$$

2. (AC 8a.2 Exercise 7) Recall that we found in previous work that $\sin(x) \approx x-\frac{1}{3!}x^3+\frac{1}{5!}x^5-\frac{1}{7!}x^7$, which is the degree 7 Taylor approximation centered at 0. And in another activity, we found that the degree 6 Taylor approximation centered at $a=0$ for $\cos(x)$ is $\cos(x) \approx 1-\frac{1}{2!}x^2+\frac{1}{4!}x^4-\frac{1}{6!}x^6$. In this exercise, we investigate Taylor polynomial approximations of $f(x)=\sin(x)$ centered at $a=\frac{\pi}{2}$.

(a) By finding the appropriate derivatives of $f(x)=\sin(x)$ and evaluating them at $a=\frac{\pi}{2}$, determine the degree 6 Taylor polynomial approximation of $\sin(x)$ centered at $a=\frac{\pi}{2}$.

$$\begin{aligned} f(x) &= \sin(x) & f(\tfrac{\pi}{2}) &= 1 \\ f'(x) &= \cos(x) & f'(\tfrac{\pi}{2}) &= 0 \\ f''(x) &= -\sin(x) & f''(\tfrac{\pi}{2}) &= -1 \\ f'''(x) &= -\cos(x) & f'''(\tfrac{\pi}{2}) &= 0 \\ f^{(4)}(x) &= \sin(x) & f^{(4)}(\tfrac{\pi}{2}) &= 1 \\ f^{(5)}(x) &= \cos(x) & f^{(5)}(\tfrac{\pi}{2}) &= 0 \\ f^{(6)}(x) &= -\sin(x) & f^{(6)}(\tfrac{\pi}{2}) &= -1 \end{aligned}$$

So then, the degree 6 approximation centered at $a = \frac{\pi}{2}$ is $\sin(x) \approx 1 - \frac{1}{2!}\left(x - \frac{\pi}{2}\right)^2 + \frac{1}{4!}\left(x - \frac{\pi}{2}\right)^4 - \frac{1}{6!}\left(x - \frac{\pi}{2}\right)^6$.

(b) How is your result in (a) similar to the degree 6 Taylor polynomial of $\cos(x)$ that is centered at $a = 0$?

Well hey, it's totally the same, except all of the $x$'s have been replaced by $\left(x - \frac{\pi}{2}\right)$'s.

(c) Recall the trigonometric identity that states $\sin(x) = \cos(x - \frac{\pi}{2})$. How does this identity help you explain what you found in (a) and (b)?

This is actually a perfect description of what has happened: we took the thing we got for $\cos(x)$ and just replaced $x$ by $(x - \frac{\pi}{2})$, and according to this trigonometric identity, that gives us the values of $\sin(x)$.

3. (AC 8a.2 Exercise 8) In earlier work, we found that

$$f(x) = \ln(1+x) \approx T_5(x) = 1x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4 + \frac{1}{5}x^5,$$

where $T_5$ is the degree 5 Taylor approximation of $\ln(1+x)$ centered at $a = 0$.

We've also found that

$$g(x) = \ln(x) \approx P_5(x) = 1(x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \frac{1}{5}(x-1)^5,$$

where $P_5$ is the degree 5 Taylor approximation of $\ln(x)$ centered at $a = 1$. (Here we are using "$T_5$" and "$P_5$" to distinguish between these two degree 5 polynomial approximations of the two different functions $f(x) = \ln(1+x)$ and $g(x) = \ln(x)$, centered at two different values.

I've punched both of these polynomial approximations into Desmos in this calculator.

(a) Note that $f(0.5) = \ln(1.5)$. Use $T_5(x)$ appropriately to find an estimate of $\ln(1.5)$.

Since $\ln(1.5) = f(0.5) \approx T_5(0.5)$, the estimate provided by $T_5(x)$ is $\ln(1.5) \approx T_5(0.5) = .40729166666\ldots$.

(b) Observe that $g(1.5) = \ln(1.5)$. Use $P_5(x)$ to estimate $\ln(1.5)$.

Since $\ln(1.5) = g(1.5) \approx P_5(1.5)$, the estimate provided by $P_5(x)$ is $\ln(1.5) \approx P_5(1.5) = .40729166666\ldots$.

(c) Are the estimates of $\ln(1.5)$ generated by $T_5(x)$ and $P_5(x)$ the same or different? Why do you think this happens?

They're the same! Since $g(x)$ and $f(x)$ are just horizontal shifts of each other, their Taylor polynomials should agree as long as we compensate for the horizontal shift by recentering the approximation.